



Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.

UNIT 5 • STANDARD 6.GR.2

Volume of Rectangular Prisms

Lesson 5-10 • Teacher Copy

Name: _____

Date: _____

Class: _____



TODAY'S OBJECTIVES

Content Objective: I can find the volume of a rectangular prism, including ones with fractional edge lengths, using $\text{base area} \times \text{height}$.

Language Objective: I can explain my work using the words volume, rectangular prism, dimensions, and base area.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Volume

Rectangular prism

Cubic units

Dimensions

Base area



Tap any word to see its meaning and a picture.

1

The amount of space inside a three-dimensional solid is its

2

A solid with six rectangular faces is a .

3

The unit used to measure volume is .

4

The measurements of length, width, and height of a solid are its

5

The area of the bottom face of a prism, multiplied by height to find volume, is the

Write About the Math

How many gallons of water does the small tank hold?

¿Cuántos galones de agua tiene la pecera pequeña?

Say your answer to a partner first. Then write it, using at least two of these words:

☐ **Volume**
Volumen

☐ **Rectangular prism**
Prisma rectangular

☐ **Cubic units**
Unidades cúbicas

☐ **Dimensions**
Dimensiones

☐ **Base area**
Área de la base

Model: You still multiply $\text{length} \times \text{width} \times \text{height}$ even when an edge is a fraction or decimal. — Students explain the formula stays the same ($V = l \times w \times h = 2 \times 1.5 \times 1 = 3 \text{ ft}^3$) and that a fractional edge just means multiplying with a fraction or decimal.

Start

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **The tank's volume is ____ in³ because $V = \text{____} \times \text{____} \times \text{____}$. It holds about ____ gallons because ____ $\div 231 \approx \text{____}$.**

- **My answer is ____ because ____.**

Mi respuesta es ____ porque ____.

Build

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**

Mi afirmación es ____.

- **My evidence is ____, so ____.**

Mi evidencia es ____, entonces ____.

Explain

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

☐ I answered the question.

☐ I used math evidence: numbers, an equation, or a model.

☐ I used at least two math words.

Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.
- The support level changed the language scaffolding, never the mathematical expectation.

Answer Key & Teacher Guide

1. **Notes 1:** Volume
2. **Notes 2:** Rectangular prism
3. **Notes 3:** Cubic units
4. **Notes 4:** Dimensions
5. **Notes 5:** Base area
6. **Watch for this mistake:** *A common mistake in Volume of Rectangular Prisms is rounding a fractional edge length to the nearest whole number before multiplying, instead of using the exact fraction or decimal. For a box that is $2\text{ ft} \times 1.5\text{ ft} \times 1\text{ ft}$, a student might round 1.5 down to 1 and compute $2 \times 1 \times 1 = 2\text{ ft}^3$, but the correct volume uses the exact edge length: $2 \times 1.5 \times 1 = 3\text{ ft}^3$ — a full cubic foot larger. Before you submit, ask: 'Did I multiply using the exact fractional or decimal edge length, or did I round it off first?'*
7. **Try It 1:** B. Box A, because 50 cubic in is more than 48 cubic in — Box A: $5 \times 5 \times 2 = 50$ cubic in. Box B: $4 \times 4 \times 3 = 48$ cubic in. $50 > 48$, so Box A holds more.
8. **Try It 2:** C. 18 cubic in — $V = 4 \times 3 \times 1.5 = 12 \times 1.5 = 18$ cubic in. A fractional edge does not change the steps.